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Inventor(s)

Odnoblyudov; Vladimir et al.

Solid state lighting devices with improved current spreading and light extraction and associated methods

Abstract

Solid state lighting (“SSL”) devices with improved current spreading and light extraction and associated methods are disclosed herein. In one embodiment, an SSL device includes a solid state emitter (“SSE”) that has a first semiconductor material, a second semiconductor material spaced apart from the first semiconductor material, and an active region between the first and second semiconductor materials. The SSL device can further include a first contact on the first semiconductor material and a second contact on the second semiconductor material and opposite the first contact. The second contact can include one or more interconnected fingers. Additionally, the SSL device can include an insulative feature extending from the first contact at least partially into the first semiconductor material. The insulative feature can be substantially aligned with the second contact.

Inventors: Odnoblyudov; Vladimir (Eagle, ID), Schubert; Martin F. (Mountain View, CA)

Applicant: Micron Technology, Inc. (Boise, ID)

Family ID: 1000008781428

Assignee: Micron Technology, Inc. (Boise, ID)

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Primary Examiner: Nadav; Ori

Attorney, Agent or Firm: Dorsey & Whitney LLP

Background/Summary

RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 16/229,151, filed Dec. 21, 2018, which is a divisional of U.S. patent application Ser. No. 13/165,248, filed Jun. 21, 2011, now U.S. Pat. No. 10,170,668, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The disclosed embodiments relate to solid state lighting (“SSL”) devices and methods of manufacturing SSL devices. In particular, the disclosed embodiments are related to SSL devices with improved current spreading and light extraction, and associated methods of use and manufacture.

BACKGROUND

(2) Mobile phones, personal digital assistants (“PDAs”), digital cameras, MP3 players, and other portable electronic devices utilize SSL devices (e.g., light emitting diodes (“LEDs”)) for background illumination. SSL devices are also used for signage, indoor lighting, outdoor lighting, and other types of general illumination. FIGS. 1A and 1B are cross-sectional and plan views, respectively, of a conventional SSL device **10** including an LED structure **11** having N-type gallium nitride (GaN) **14**, GaN/indium gallium nitride (InGaN) multiple quantum wells (“MQWs”) **16**, and P-type GaN **18**. The SSL device **10** also includes a first contact **20** at the P-type GaN **18** and a second contact **22** at the N-type GaN **14**. The first contact **20** includes a sheet-like structure across the underside of the P-type GaN **18**. The second contact **22** includes a plurality of fingers **21** (three are shown for illustration purposes) coupled to one another by a cross member **23** (FIG. 1B). In operation, a continuous or pulsed electrical voltage is applied between the first and second contacts **20** and **22**. In response, as electrical current flows from the second contact **22**, through the N-type GaN **14**, the GaN/InGaN MQWs **16**, and the P-type GaN **18** to the first contact **22**, the GaN/InGaN MQWs **16** convert a portion of the electrical energy into light. The generated light is extracted from the N-type GaN **14** of the SSL device **10** for illumination and/or other suitable purposes.

(3) SSL devices having contact structures on the front and back surface of the LED structure **11**, such as the fingers **21** and the first contact **20** shown in FIGS. 1A and 1B, are known to those skilled in the art as “thin-film” SSL devices. Such thin-film SSL devices have high thermal performance and efficiency. The fingers **21**, however, may be opaque and thus block some of the light. For example, according to conventional techniques, the first and second contacts **20** and **22** typically include aluminum, copper, or other opaque conductive materials. As a result, the second contact **22** blocks light generated in areas directly beneath it. These same areas, however, have the highest current density across the SSL device **10** and thus produce the highest intensity of light in the SSL device **10**. Consequently, regions with the highest light intensity per unit area also have the lowest probability of extracting light, which reduces light extraction efficiencies. Accordingly, several improvements in increasing current spreading and light extraction efficiency in SSL devices may be desirable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic cross-sectional diagram of an SSL device in accordance with the prior art.

(2) FIG. 1B is a schematic plan view of the SSL device in FIG. 1A.

(3) FIG. 2A is a schematic cross-sectional view of an SSL device in accordance with an embodiment of the present technology.

(4) FIG. 2B is a schematic top plan view of the SSL device shown in FIG. 2A.

(5) FIG. 2C is a schematic plan view of a portion of the SSL device taken substantially along line 2C-2C of FIG. 2A.

(6) FIGS. 3A-3E are schematic cross-sectional views of a method of fabricating SSL devices in accordance with embodiments of the present technology.

(7) FIGS. 4A-4D are schematic cross-sectional views of a method of fabricating SSL devices in accordance with another embodiment of the present technology.

- (8) FIG. 5 is a schematic cross-sectional view of an SSL device in accordance with a further embodiment of the present technology.
- (9) FIG. 6 is a schematic cross-sectional view of an SSL device in accordance with an additional embodiment of the present technology.
- (10) FIG. 7 is a schematic cross-sectional view of an SSL device in accordance with yet another embodiment of the present technology.
- (11) FIG. 8 is a schematic cross-sectional view of an SSL device in accordance with still another embodiment of the present technology.

DETAILED DESCRIPTION

(12) Specific details of several embodiments of SSL devices with improved current spreading and light extraction and associated methods of using and manufacturing such SSL devices are described below. The term “SSL” generally refers to “solid state light” and/or “solid state lighting” according to the context in which it is used. The term solid state emitter (“SSE”) generally refers to solid state light emitting dies that convert electrical energy into electromagnetic radiation in the visible, ultraviolet, infrared, and/or other spectra. SSEs include semiconductor light-emitting diodes (“LEDs”), polymer light-emitting diodes (“PLEDs”), organic light-emitting diodes (“OLEDs”), or other types of solid state devices that convert electrical energy into electromagnetic radiation in a desired spectrum. The term “light extraction efficiency” generally refers to a ratio of the amount of light extracted from an SSL device to the total light generated in the SSL device. A person skilled in the relevant art will also understand that the technology may have additional embodiments, and that the technology may be practiced without several of the details of the embodiments described below with reference to FIGS. 2A-8.

(13) FIG. 2A is a schematic cross-sectional view of an SSL device **200** in accordance with an embodiment of the technology. FIG. 2B is a top plan view of the SSL device **200** of FIG. 2A, and FIG. 2C is a plan view of a portion of the SSL device **200** taken substantially along line 2C-2C of FIG. 2A. As shown in FIG. 2A, the SSL device **200** can include a support **202** and an SSE **204** attached to the support **202**. The SSL device **200** can further include a first contact **216** at a first side **206a** of the SSE **204** and a second contact **218** at a second side **206b** of the SSE **204** opposite the first side **206a**. Additionally, the SSL device **200** can include an insulative feature **214** at the first side of the SSE **204**.

(14) As shown in FIG. 2A, the SSE **204** can include a first semiconductor material **208**, an active region **210**, and a second semiconductor material **212**. The first semiconductor material **208** can be a P-type semiconductor material proximate a first side **206a** of the SSE **204**, such as P-type gallium nitride (“P-GaN”). The second semiconductor material **212** can be an N-type semiconductor material proximate a second side **206b** of the SSE **204**, such as N-type gallium nitride (“N-GaN”). This configuration is suitable for SSEs **204** formed on opaque or translucent growth substrates and subsequently attached to the support **202**. In selected embodiments, the first and second semiconductor materials **208** and **212** can individually include at least one of gallium arsenide (GaAs), aluminum gallium arsenide (AlGaAs), gallium arsenide phosphide (GaAsP), gallium (III) phosphide (GaP), zinc selenide (ZnSe), boron nitride (BN), aluminum gallium nitride (AlGaN), and/or other suitable semiconductor materials.

(15) The active region **210** can include a single quantum well (“SQW”), MQWs, and/or a bulk semiconductor material. The term “bulk semiconductor material” generally refers to a single grain semiconductor material (e.g., InGaN) with a thickness between approximately 10 nanometers and approximately 500 nanometers. In certain embodiments, the active region **210** can include an InGaN SQW, GaN/InGaN MQWs, and/or an InGaN bulk material. In other embodiments, the active region **210** can include aluminum gallium indium phosphide (AlGaInP), aluminum gallium indium nitride (AlGaInN), and/or other suitable materials or configurations.

(16) As further shown in FIG. 2A, the first contact **216** can be electrically coupled to the first semiconductor material **208**, and the second contact **218** can be electrically coupled to the second

semiconductor material **212**. The first contact **216** can be a layer of copper (Cu), aluminum (Al), tungsten (W), stainless steel, silver (Ag), nickel (Ni), and/or other suitable metal or metal alloy deposited on the first semiconductor material **208**. The second contact **218** can have a structure generally similar to the first contact **20** shown in FIG. **1B**. For example, as shown in FIG. **2B**, the second contact **218** can include a plurality of fingers **220** connected to one another by a cross member **222**. The fingers **220** and/or the cross member **222** can individually include an elongated structure as shown in FIG. **2B** and/or other suitable structures. The fingers **220** and/or the cross member **222** can include copper (Cu), aluminum (Al), silver (Ag), gold (Au), platinum (Pt), and/or other suitable electrically conductive materials. In other embodiments, the second contact **218** can be a transparent electrode constructed from indium tin oxide (“ITO”), aluminum zinc oxide (“AZO”), fluorine-doped tin oxide (“FTO”), and/or other substantially transparent and conductive oxides.

(17) The insulative feature **214** is configured to prevent ohmic contact at selected areas of the SSE **204**. For example, as shown in FIG. **2A**, portions of the insulative feature **214** can be substantially aligned (e.g., superimposed) with the fingers **220** to interrupt or block electrical current flowing directly between the first and second contacts **216** and **218**. In the embodiment illustrated in FIG. **2C**, for example, the insulative feature **214** includes three fingers **228** that correspond to the three fingers **220** of the second contact **218** (FIG. **2B**) and a cross member **230** that corresponds to the cross member **222** of the second contact **218**. In other embodiments, the insulative feature **214** can have other configurations that do not necessarily correspond to the size, number, or shape of the features of the second contact **218**.

(18) In the embodiment illustrated in FIG. **2A**, the insulative feature **214** extends through the active region **210** and has a substantially equal width at the first side **206a** of the SSE **204** as the width of the fingers **220** of the second contact **218**. This configuration substantially eliminates the current density directly between the first and second contacts **216** and **218**, and thereby substantially blocks light production under the second contact **218**. In other embodiments, the insulative feature **214** can extend varying depths into the first semiconductor material **208**, the active region **210**, and/or the second semiconductor material **212** to vary the current density under the second contact **218**. In further embodiments, the insulative feature **214** can have a width that is greater than or less than that of the corresponding portions of the second contact **218**. In any of the foregoing embodiments, the protrusion of the insulative feature **214** into the SSE **204** and the width of the insulative feature **214** may be selected based on a target current flow profile in the SSL device **200**.

(19) The insulative feature **214** can include a dielectric material **224** to electrically insulate the first contact **216**. The dielectric material **224** can be a substantially conformal layer as shown in FIG. **2A** that includes silicon oxide, silicon nitride, magnesium oxide, and/or other suitable dielectric materials. In other embodiments, the dielectric material **224** can comprise a larger portion of the insulative feature **214**. In further embodiments, the insulative feature **214** can include other electrically insulative materials and/or electrically insulative structures (e.g., air pockets).

(20) As shown in FIG. **2A**, the SSL device can further include a reflective material **226** that covers the backside of the first contact **216** and the insulative feature **214**. When the insulative feature **214** is transparent (e.g., made of a transparent dielectric material **224**), the reflective material **226** reflects and scatters light outboard of the second contact **218**. The reflective material **226** can include gold (Au), copper (Cu), silver (Ag), aluminum (Al), and/or any other suitable material that reflects light emitted from the active region **210**. The reflective material **226** can be selected based on its thermal conductivity and/or the color of light it reflects. For example, silver generally does not alter the color of the reflected light. Gold, copper, or other colored reflective materials can affect the color of the light and can accordingly be selected to produce a desired color for the light emitted by the SSL device **200**. The insulative feature **214** and the reflective material **226** are configured to prevent light from being obstructed by the second contact **218** and further increase the light extraction efficiency of the SSL device **200**.

(21) When the reflective material **226** covers the backside of the first contact **216**, the reflective layer **226** can provide additional adhesion between the first contact **216** and the first semiconductor material **208** to prevent electrical shorting and improve the reliability of the SSL device **200**. This configuration can be especially advantageous when the first contact **216** is silver-based because it can increase the adhesion strength of the silver and prevent silver electromigration.

(22) Several embodiments of the SSL device **200** shown in FIGS. 2A-2C can have enhanced current spreading and a higher light extraction efficiency than conventional SSL devices. As explained above with reference to FIGS. 1A and 1B, conventional SSL devices are inefficient because, among other reasons, the highest current density in conventional SSL devices is directly under the second contacts. The SSL device **200** avoids this problem because the insulative feature **214** shown in FIGS. 2A and 2C reduces or eliminates the current density directly under the second contact **218**. This spreads the current into the non-contact areas that are not blocked by the second contact **218** and causes the non-contact areas to have higher current densities than without the insulative feature **214**. The SSL device **200**, therefore, has a high extraction efficiency because more light is generated from the non-contact areas where the light can be readily extracted without being blocked by the second contact **218**. Additionally, the insulative feature **214** and the reflective material **226** shown in FIGS. 2A and 2C can scatter light and further increase the light extraction efficiency of the SSL device **200**.

(23) FIGS. 3A-3E illustrate a process of forming the SSL device **200** in accordance with an embodiment of the present technology. FIG. 3A shows a stage of the process after the SSE **204** has been formed on a formation substrate **332**. The SSE **204** can be formed via metal organic chemical vapor deposition (“MOCVD”), molecular beam epitaxy (“MBE”), liquid phase epitaxy (“LPE”), and/or hydride vapor phase epitaxy (“HVPE”). In other embodiments, at least one of the first semiconductor material **208**, the active region **210**, and the second semiconductor material **212** may be formed via other suitable epitaxial growth techniques. Additionally, as shown in FIG. 3A, the first contact **216** can be formed over the first semiconductor material **208** using chemical vapor deposition (“CVD”), physical vapor deposition (“PVD”), atomic layer deposition (“ALD”), or other techniques known in the semiconductor fabrication arts. A masking material **334** (e.g., a photoresist) can be deposited and processed to have a desired pattern of openings **335** over the first contact **216**.

(24) FIG. 3B illustrates a stage of the process after etching one or more trenches **336** through the first contact **216** and into at least a portion of the SSE **204**. In the embodiment shown in FIG. 3B, for example, the trenches **336** extend through the junction between the first and second semiconductor materials **208** and **212**. In other embodiments, the trenches **336** can extend deeper or shallower into the SSE **204** and/or be formed using other suitable semiconductor fabrication techniques. The trenches **336** can be V-shaped as shown in FIG. 3B, and/or they can have other suitable shapes. In selected embodiments, the shapes, positions, and dimensions of the trenches **336** are based on the desired shape, position, and dimension of the subsequently formed second contact **218** (FIG. 3E). For example, the trenches **336** can be configured such that they define the configuration of the fingers **228** and the cross member **230** described above with reference to FIGS. 2A-2C. After forming the trenches **336**, the masking material **334** is removed from the remaining portions of the first contact **216**.

(25) FIG. 3C shows a stage of the process after depositing the dielectric material **224** into the trenches **336** and over the remaining portions of the first contact **216**. The dielectric material **224** can be deposited using CVD, ALD, spin coating, or other suitable deposition methods. The dielectric material **224** can be a conformal layer as shown in FIG. 3C. In other embodiments, the dielectric material **224** can be deposited such that it substantially fills the trenches **336**. The overburden of the dielectric material **224** that covers the first contact **216** (e.g., line **338**) can be removed using dry etching, wet etching, laser ablation, and/or other suitable material removal techniques.

(26) FIG. 3D illustrates a stage of the process after the reflective material **226** has been deposited over the dielectric material **224** and the first contact **216**. The reflective material **226** can be deposited using CVD, ALD, plating, and/or other suitable semiconductor fabrication techniques. A portion of the reflective material **226** can be removed (e.g., along line **340**) using dry etching, wet etching, laser ablation, grinding, planarizing, and/or other suitable material removal techniques. In the embodiment shown in FIG. 3D, a portion of the reflective material **226** remains on the first contact **216** such that the reflective material **226** effectively encapsulates the first contact **216** to strengthen the bond between the first contact **216** and the first semiconductor material **208**.

(27) FIG. 3E illustrates a stage of the process after the formation substrate **332** has been removed from the second semiconductor material **212** and the reflective material **226** has been attached to the support **202** via a bonding process (not shown). The process can further include forming the second contact **218** on the second semiconductor material **212**. As shown in FIG. 3E, the second contact **218** can correspond to the insulative feature **214** such that the current density under the second contact **218** is substantially reduced or eliminated. Optionally, as shown in FIG. 3E, the process can also include singulating individual SSL devices **200** along lines **337-337**. The process may also optionally include surface roughening for increased light extraction.

(28) FIGS. 4A-4D illustrate a process of forming an SSL device **400** in accordance with another embodiment of the present technology. The process can include generally similar steps as the process described above with reference to FIGS. 3A-3E. For example, FIG. 4A shows a stage of the process after the trenches **336** have been formed in the SSE **204**. FIG. 4A is similar to FIGS. 3A and 3B, but in FIG. 4A the trenches **336** are formed before depositing a conductive material for the first contact **216** onto the first semiconductor material **208**. In the embodiment shown in FIG. 4A, the masking material **334** (e.g., a photoresist) is accordingly formed on the first semiconductor material **208** in a desired configuration such that portions of the SSE **204** can be removed to form the trenches **336**.

(29) As shown in FIGS. 4B and 4C, the process can further include forming the dielectric material **224** in the trenches **336** (FIG. 4B) and depositing the reflective material **226** over the dielectric material **224** (FIG. 4C). Referring to FIG. 4B, the overburden of the dielectric material **224** can be removed along line **442** to expose the portions of the first semiconductor material **208** under the dielectric material **224**. Referring to FIG. 4C, the reflective material **226** can be conductive and electrically coupled to the first semiconductor material **208**. The reflective material **226** can thus define a first contact **416**. In this configuration, the reflective material **226** can also redirect light back through the first semiconductor material **208**, the active region **210**, and the second semiconductor material **212**.

(30) FIG. 4D shows a subsequent stage of the process after the SSE **204** has been removed from the formation substrate **332** and the reflective material **226** has been attached to the support **202**. The second contact **218** can then be formed on the second semiconductor material **212** such that it generally corresponds to at least a portion of the insulative feature **214**. The insulative feature **214** can reduce or eliminate current from flowing directly under the second contact **218** and thereby increase the light extraction of the SSL devices **400** in a manner similar to the SSL devices **200** described above.

(31) FIG. 5 is a schematic cross-sectional view of an SSL device **500** in accordance with another embodiment of the present technology. Several features of the SSL device **500** are generally similar to the features of the SSL devices **200** and **400** described above with reference to FIGS. 2A-4D. For example, the SSL device **500** includes the support **202**, the SSE **204**, and the first and second contacts **216** and **218** spaced vertically apart from one another. Additionally, as shown in FIG. 5, the SSL device **500** includes the insulative feature **214** that is generally aligned with the second contact **218** to enhance the light extraction efficiency of the SSL device **500**. However, rather than the insulative feature **214** including a dielectric material and/or a reflective material as described above, the insulative feature **214** illustrated in FIG. 5 includes one or more air-filled gaps that can

spread the current in the SSE **204** and scatter light generated by the SSE **204**. In other embodiments, the insulative feature **214** can include another suitable insulative gas. Additionally, as shown in FIG. 5, the insulative feature **214** can also have a different shape than the V-shape shown in FIGS. 2A-4D and different dimensions than the second contact **218**.

(32) FIG. 6 is a schematic cross-sectional view of an SSL device **600** in accordance with an additional embodiment of the present technology. Several features of the SSL device **600** are generally similar to the features of the SSL devices **200**, **400** and **500** described above with reference to FIGS. 2A-5. As shown in FIG. 6, the SSL device **600** can include a first insulative feature **214a** and a second insulative feature **214b**. Similar to the insulative feature **214** shown in FIG. 5, the first insulative feature **214a** can include one or more air-filled gaps that are substantially aligned with the second contact **218** to spread current and scatter light from under the second contact **218**. In the embodiment illustrated in FIG. 6, the first insulative feature **214a** extends only partially into the first semiconductor material **208** and therefore reduces, rather than eliminates, the current density under the second contact **218**. Additionally, as shown in FIG. 6, the first insulative feature **214a** can also vary in width from the insulative features **214** shown in FIGS. 2A-5.

(33) The second insulative feature **214b** can be positioned at non-contact portions of the SSE **204**. Similar to the insulative features **214** described above with reference to FIGS. 2A-4D, the second insulative feature **214b** can include the dielectric material **224** that extends through the junction between the first and second semiconductor materials **208** and **212**. The second insulative feature **214b** can further control current spreading and scatter light.

(34) FIG. 7 is a schematic cross-sectional view of an SSL device **700** in accordance with yet another embodiment of the present technology. Several features of the SSL device **700** are generally similar to the features of the SSL devices **200**, **400**, **500** and **600** described above with reference to FIGS. 2A-6. However, rather than the first and second contacts **216** and **218** being spaced vertically apart from one another as described above, the first and second contacts **216** and **218** illustrated in FIG. 7 are spaced laterally apart from one another. Like the vertical SSL devices described above, the lateral SSL device **700** shown in FIG. 7 can also benefit from increased light extraction efficiencies provided by the insulative feature **214**.

(35) FIG. 8 is a schematic cross-sectional view of an SSL device **800** configured in accordance with yet another embodiment of the present technology. Several features of the SSL device **800** are generally similar to the features of the SSL devices **200**, **400**, **500** and **600** described above with reference to FIGS. 2A-6. The SSL device **800** can include an insulative feature **814** positioned between portions of the first contact **216** and the first semiconductor material **208**, rather than extending into the SSE **204** like the insulative features **214** described above. However, similar to the insulative features **214** described above, the insulative features **814** shown in FIG. 8 can be positioned where ohmic contact is not desirable (e.g., aligned with the second contact **218**) to reduce the current density under the second contact **218**. The insulative feature **814** shown in FIG. 8 can be filled with a dielectric material (e.g., the dielectric material **224** described above), an insulative gas (e.g., air), and/or other insulative material.

(36) From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but that various modifications may be made without deviating from the disclosure. For example, in FIGS. 2A-6 and 8, the first contact **216** is a blanket contact that covers the backside of the first semiconductor material **208**. In other embodiments, the first contact **216** can be patterned such that it includes discrete portions and/or have other suitable contact configurations. Additionally, in any of the embodiments illustrated, the SSL devices can optionally include an optical component (e.g., a collimator, a lens) and/or other suitable components. Certain aspects of the new technology described in the context of particular embodiments may be combined or eliminated in other embodiments. For example, any of the SSL devices described above can include more than one insulative feature and/or portions of the insulative features can include different structures (e.g., air-filled gaps, dielectric material, etc.).

Additionally, although advantages associated with certain embodiments of the new technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1. A method of forming a solid state lighting device, the method comprising: forming a solid state emitter (SSE) having a first side and a second side opposite the first side; forming a first trench at the first side of the SSE, the first trench extending at least partially into the SSE; depositing an insulative material in the first trench to form an insulative feature, wherein the insulative material is conformal to the first trench and defines a second trench within the first trench; depositing a reflective material over the insulative material, wherein the reflective material completely fills the second trench; planarizing the reflective material such that a continuous planar surface of the reflective material overlies the insulative feature; and forming a contact on the second side of the SSE, the contact having a conductive finger vertically aligned with the first trench.
2. The method of claim 1 wherein depositing the insulative material comprises depositing a dielectric material into the first trench.
3. The method of claim 1 wherein depositing the insulative material comprises filling the trench with air.
4. The method of claim 1 wherein: forming the SSE comprises forming a first semiconductor material, an active region, and a second semiconductor material sequentially on one another, the first semiconductor material being at the first side of the SSE and the second semiconductor material being at the second side of the SSE; and forming the first trench comprises extending the first trench through the active region.
5. The method of claim 1 wherein: forming the first contact comprises forming a first-side contact before forming the first trench; and forming the first trench comprises forming the first trench through the first-side contact.
6. The method of claim 1 wherein: forming the trench comprises forming the trench before forming a first-side contact; depositing the insulative material comprises depositing a dielectric material into the trench; and forming the first-side contact comprises forming a reflective material over the dielectric material and over the first side of the SSE.
7. The method of claim 1 wherein the insulative feature is a first insulative feature, and wherein the method further comprises: forming a second insulative feature at the first side of the SSE, the second insulative feature being spaced laterally apart from the contact and extending at least partially into the SSE.
8. The method of claim 1 wherein the SSE includes a non-contact area outboard the area beneath the contact, and wherein the insulative feature is configured to increase current spreading in the non-contact area.
9. The method of claim 1, wherein: the insulative material partially fills the first trench; and the reflective material fills the second trench from a backside of the insulative material past a surface facing away from the first side of the SSE.
10. A method of forming a solid state lighting device, the method comprising: forming a solid state emitter (SSE) having a first side and a second side opposite the first side; forming an insulative feature having insulative fingers with first trenches extending at least partially into the SSE, wherein the insulative feature includes a transparent dielectric material having a backside defining second trenches within the first trenches and a reflective material deposited at the backside of the transparent dielectric material, the reflective material completely filling the second trenches; planarizing the reflective material such that a continuous planar surface of the reflective material

overlies the transparent dielectric material; and forming a contact at the second side of the SSE, wherein forming the contact includes vertically aligning conductive fingers of the contact with corresponding ones of the insulative fingers.

11. The method of claim 10 wherein: forming the SSE comprises forming a first semiconductor material, an active region, and a second semiconductor material sequentially on one another, the first semiconductor material being at the first side of the SSE and the second semiconductor material being at the second side of the SSE; and forming the insulative feature includes forming the insulative fingers to extend through the first semiconductor material and at least partially into the active region of the SSE.

12. The method of claim 10 wherein: forming the insulative feature includes forming an insulative cross member interconnecting the insulative fingers; and forming the contact includes forming a conductive cross member interconnecting the conductive fingers.

13. The method of claim 10 wherein the SSE includes non-contact areas outboard the area beneath the conductive fingers, and wherein forming the contact includes configuring the contact to increase current spreading in the non-contact areas.

14. The method of claim 10, wherein: the transparent dielectric material partially fills the first trenches; and the reflective material fills the second trenches from the backside of the transparent dielectric material past a surface facing away from the first side of the SSE.

15. The method of claim 10 wherein the insulative feature at least partially comprises a transparent dielectric material.

16. A method of operating a solid state lighting device including a solid state emitter (SSE) having a first contact on a first surface thereof and a second contact on a second surface thereof, the method comprising: flowing current between the first and second contacts to produce light for emission through the second surface of the SSE; and reducing the current density under the second contact to substantially block light production under the second contact, wherein the solid state light device includes an insulative feature having a plurality of insulative fingers with first trenches extending from the first contact at least partially into the SSE, wherein the second contact includes a plurality of conductive fingers, wherein each of the conductive fingers is vertically aligned with a corresponding one of the insulative fingers, and wherein the insulative feature includes a transparent dielectric material having a backside defining second trenches within the first trenches and a reflective material deposited at the backside of the transparent dielectric material, the reflective material completely filling the second trenches and having a continuous planar surface that overlies the insulative feature.

17. The method of claim 16 wherein reducing the current density under the second contact includes substantially eliminating a direct current path orthogonal to the second surface of the SSE between the first and second contacts.

18. The method of claim 17 wherein the method further comprises reflecting light from the first surface toward the second surface.

19. The method of claim 16, wherein: the transparent dielectric material partially fills the first trenches; and the reflective material fills the second trenches from the backside of the transparent dielectric material past a plane coplanar with a side of the first contact, the side facing away from the first surface of the SSE.
